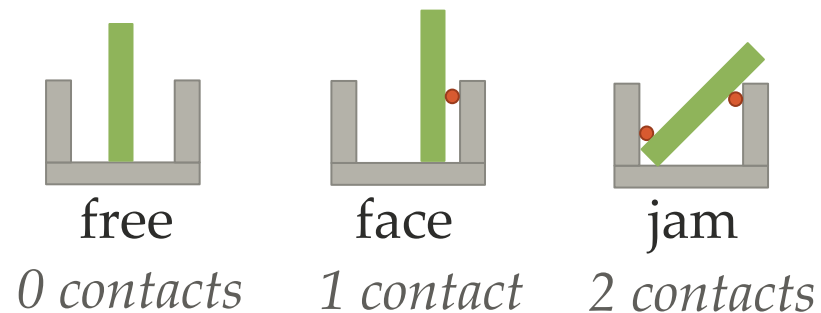
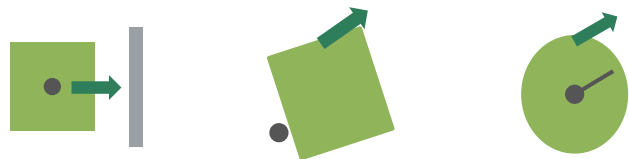


(a) Example Hybrid Action Tasks

Contact-rich Assembly



Articulated Object Manipulation



slide
prismatic

rotate
revolute

twist
axial

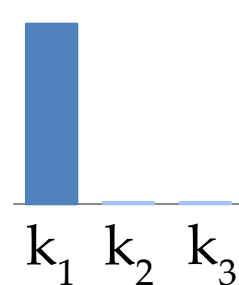
Discrete mode + continuous parameters

■ *fixture / wall* ■ *movable part* ● *joint axis*
● *contact point* action $a = (m, x)$

(b) Diffusion vs BFN on Hybrid Actions

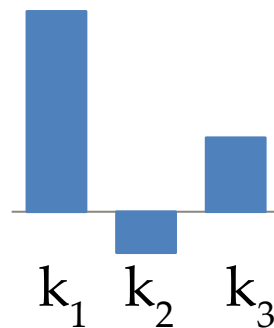
Diffusion: Relaxes Discrete to Continuous

one-hot target



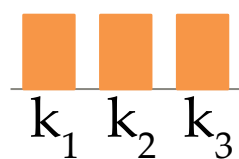
+ Gaussian
noise

noisy output



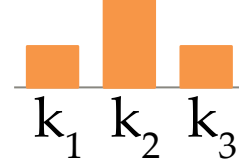
Invalid!

BFN: Refines a Categorical Belief on the Simplex



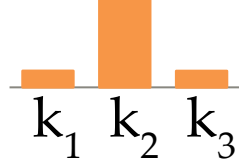
$t=0$

uniform



$t=1/2$

peaked



$t=1$

peaked

Valid!

(c) Headline Results

Success Rate

